

From 590 papers to 58

How the ARC-AGI-3 literature review was actually run — the funnel, the three rules that were overturned by their own output, and what the extraction settled.

Compiled 5 September 2026 **Deadline** 30 September 2026 **Coverage** 56 of 58 (96.6%)
Method snowball + hand triage · not PRISMA

The funnel

Eleven seed papers, one citation hop in each direction, then a hand triage down to a set small enough to read. Every number below is reproducible from the scripts in `research_prep/scripts/`.

11 Seed papers 6 RECENT · 5 CLASSICAL	512 Backward references SEMANTIC SCHOLAR API	83 Forward citations 8 LISTS · HAND- TRIAGED	590 Merged pool, deduped NORMALISED TITLE KEY
58 Shortlist QUOTA BY CLUSTER	56 PDFs extracted 2 UNREACHABLE		

The two missing papers are acquisition failures — one behind an IEEE paywall, one with no DOI — not screening decisions. That distinction belongs in any write-up: a reader who later finds either paper should find it already accounted for.

How it was run

Five stages. The order matters — stage 5 deliberately comes after stage 4, and doing it the other way round is the most common way this kind of review goes wrong.

01 Pick seeds for coverage, not quality

Six 2026 papers for the current field, five classical ones — C-SWM, DRQN, Episodic Control, Plan2Explore, World Models — to reach the mechanism literature the recent work assumes rather than cites. Together they must open a path into all five clusters.

seeds → data/candidates.csv

02 Hop both directions, and count co-citation

One backward hop per seed via the Semantic Scholar graph API, plus forward hops for papers citing each seed. The ranking signal is `seed_overlap` — how many seeds cite a given paper — which is sortable, unlike an impression from a graph view.

merge_forward.py → data/candidates_merged.csv

03 Triage to a quota, per cluster

Balance beats raw score: a review returning twenty exploration papers and nothing on memory cannot rank the hypotheses. Each paper gets a verdict — `strong`, `keep`, `borderline` — and every hand judgement is written to an overrides file keyed by normalised title, so it survives the shortlist being regenerated.

data/cluster_overrides.csv

04 Extract nine columns from full text

PDFs fetched by script, then two Elicit Extract Data tables of 25 and 31 papers. Every column carries an explicit escape hatch — “*not reported*”, “*no evidence*” — which is load-bearing: without it the model fills gaps with plausible inference rather than reporting absence.

fetch_pdfs.py → 2 tables × 9 columns

Only now, assign the final categories

With each paper's extracted mechanism visible next to its title, cluster assignment stops being a guess. Seven of 58 assignments changed — and two of those moved cluster D from one paper to four, fixing a structural thinness that a pre-extraction pass would have preserved.

```
join_elicit.py → data/shortlist_extracted.csv
```

Three rules the data overturned

These are the useful part for anyone running a similar review. Each rule was written before the data existed and did not survive contact with it. The revision log is kept in the plan document rather than being tidied away — a review that hides its corrections is not reproducible.

Ranking by citation count ranks fame, not relevance

Once the classical seeds landed, the deep-learning substrate they cite swamped everything. The shortlist came back containing *Matplotlib: A 2D Graphics Environment* in the object-centric world-models cluster, alongside LSTM, *Attention Is All You Need*, and Sutton & Barto. Every real mechanism paper had been pushed out.

Fix: a paper must match at least two cluster keywords — one is noise, since “Long Short-Term Memory” matches “memory” — and anything above 25,000 citations is excluded as infrastructure. Citations survive only as a tiebreak.

A citation graph cannot find a literature that postdates its seeds

Thirteen ARC-AGI-3 papers exist. **512 backward references and eight forward lists surfaced none of them**, because every seed either predates the benchmark or is contemporary with it, and none cite it. They were found only by searching the benchmark name directly — by hand, twice, after the harvest looked complete.

Fix: always pair the graph walk with a direct name search for the benchmark, the method and the competition. This is a known boundary of snowball sampling, not a failure of it — and the fix costs one search.

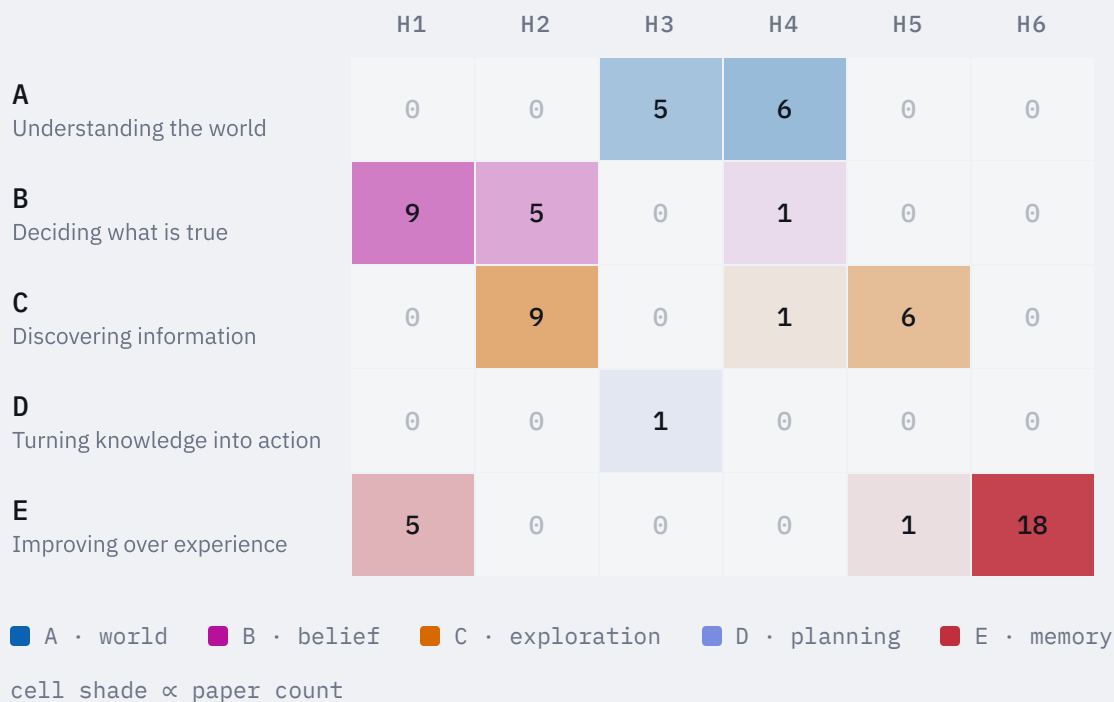
Titles are not a safe join key

The extraction tool re-derives titles from the PDF, and read the *venue* as the title for two different papers — both became “Transactions on Machine Learning”, collided on one key, and a row was silently lost. Title matching gave 21 of 25 rows with one destroyed.

Fix: join on the PDF filename, which we chose at download time and no external tool can rewrite.
Result: **56 of 56, zero unused.**

Does the filing system mean anything?

Five clusters were defined up front, six hypotheses independently. Crossing them is a validity check: if the clusters were arbitrary, the hypotheses would smear evenly across them. They do not.



What the diagonal tells us

A owns H3 and H4, B owns H1, C owns H2 and H5, E owns H6 — arrived at independently through triage, matching what the taxonomy predicted. The clusters are load-bearing, not decorative.

It also makes the thin spots unmissable. **H3 has six papers and H5 has seven** — and those are the only two hypotheses no ARC-AGI-3 paper has claimed. That coincidence is the most actionable thing on this page, though thin coverage can mean open ground or it can mean the idea does not work.

What the extraction settled

Published scores disagree by an order of magnitude — and it is protocol, not progress

RHAE is the benchmark's efficiency metric: per level, $(\text{human} \div \text{agent})^2$, capped at 1.15, level-weighted, with a per-game cap of 100. Never quote one of these numbers without its regime.

SOURCE	NUMBER	REGIME
Tycho	100.00	all 183 levels · Opus 5 / GPT-5.6 Sol
PRO-LONG	97.4%	best@2 · Fable 5 · \$1,750
Prime Agent	95.5%	best@1 · up from 30%
OPINE-World	78.4%	20/25 games · no per-game training
Rodionov ablation	34.2 – 74.8	main study · all variants and efforts
DreamTeam	38.4%	official protocol · protocol-matched · 2 runs

Rodionov, who wrote the reference architecture, states the limit himself: because the newest model postdates the games and held-out performance is untested, his results “indicate public-set saturation only.”

The gap that survives

Every published *ARC-AGI-3* result runs a hosted frontier model — confirmed across all 56 extracted papers. Kaggle's evaluation sandbox has no internet, so a submission must run local weights.

But **13 of 56 shortlisted papers ran open-weight models** in adjacent settings, so the honest claim is narrower than “nobody has tried small models”:

PAPER	WHAT IT RAN	WHY IT MATTERS
SAGE	1B – 7B, ×7	Wins on all seven backbones. Its gate resolves in closed form, with no LLM call.
Agentic Episodic Control	7B vs 32B	Grid worlds. 7B holds up, degrading most on the hardest exploration task.
PREMem	3B – 72B	A full capability ladder applied to one method.
Continual Harness	E2B – 31B	LoRA-tuned locally; a frontier teacher relabels open-weight rollouts.

So: **the capability axis has been plotted downward in memory, exploration and episodic-control settings — but never on ARC-AGI-3.** That is a better position than the original claim, because there is method to borrow rather than invent. And one design lesson is already extractable: SAGE survives at 1B *because* its decision is closed-form. Pushing work out of the model into cheap deterministic components is what makes small weights survivable.

What exists now

Everything is reproducible from the repository: re-running the scripts on the same inputs produces the same shortlist.

data/candidates.csv	512 backward references from 11 seeds, with co-citation counts
data/forward/*.csv	8 forward lists, each hand-triaged with a verdict and note
data/candidates_merged.csv	The deduped 590-paper pool
data/shortlist.csv	The 58 — regenerated by script, never hand-edited
data/cluster_overrides.csv	Every hand judgement, keyed by normalised title so it survives regeneration

data/shortlist_papers/	56 PDFs and a fetch manifest recording what could not be reached
data/shortlist_extracted.csv	58 rows × 57 columns — the shortlist joined to the extraction. This is the reading artefact.
data/implementation_refs.csv	3 MIT-licensed codebases to read before writing any agent

What is still open

BLOCKING

Kaggle account and phone verification

Gates GPU access and is the only outstanding item with an external dependency. Nothing else on this list can absorb the delay if it goes wrong.

NEXT

The ranking write-up

The deliverable this review exists to produce: H1–H6 ranked, with the evidence behind each rank, plus a named prior-art note so the build does not accidentally rebuild Tycho or AERA.

NEXT

Read the reference implementation

`arc-3-agents-baseline1` — code first, then paper — followed by Tycho, Sensi and OPINE-World. Three papers drive the build; the other 55 inform it.

OPEN

The model decision

The review narrowed it but did not settle it. A submission forces local weights, and the evidence says small open-weight models are survivable in adjacent settings. It does not name a model.

OPEN

Two unreachable papers

Excluded by decision rather than screening. Coverage is 56 of 58, and the write-up must state that plainly.

OPEN

Extend the mock, or build against the real engine?

The recommendation on paper is to skip further mock work — the mock cannot discriminate the two agent arms, and the `arc-agi` package gives local parity with the

Kaggle engine. Not formally closed.

OPEN

Efficiency ratio in `metrics.py`

Since random scores a flat RHAЕ 0.0000, win rate cannot be the headline number. Needs only a configurable reference action count, not the real interface — carried since Tuesday.

OPEN

Promote `research_prep/` into `docs/`

The staging folder was always temporary. Moves at the live build.

Method: citation-graph snowballing from 11 seeds with hand triage and full-text extraction. This is **not PRISMA**, and describing it as a systematic review would misrepresent it. Figures current to 5 September 2026 and reproducible from `research_prep/` .